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MIRAMYCIN

(Miramycin Injection 40 mg/ml (2 ml))



Name and Strength of Active Substance(s)

Gentamycin Sulfate Injection 40 mg/ml (2 ml)

Product Description

Miramycin is a clear pyrogen free solution of Gentamicin sulfate for intramuscular or intravenous injection.

Pharmacodynamics

Gentamicin sulfate is a complex mixture of the sulfate of Gentamicin C1, Gentamicin C1A and Gentamicin C2 which are produced by the microorganism *Micromonospora purpurea*. Gentamicin sulphate is a broad spectrum aminoglycoside antibiotic which is effective against a wide range of Gram-positive and Gram-negative bacteria, including *Staphylococcus aureus*, *Pseudomonas* sp. and *Proteus* sp. and many coliform bacilli. Gentamicin is taken up into sensitive bacterial cells by an active transport process which is inhibited in anaerobic, acidic or hyperosmolar environments. Within the cell they bind to the 30S and to some extent to 50S subunits of the bacterial ribosome, inhibiting protein synthesis and generating error in the transcription of the genetic code.

Pharmacokinetics

Gentamicin sulfate is poorly absorbed after oral administration but is rapidly absorbed after intramuscular administration. Average peak plasma concentrations of about 40 g per ml have been obtained in patients with normal renal function 30 to 60 minutes after intramuscular administration of a dose equivalent to 1 mg of gentamicin per kg body-weight, which is similar to concentrations achieved after intravenous infusion. Binding of gentamicin to plasma proteins is usually low. It diffuses mainly into extracellular fluids, it crosses the placenta but only small amounts have been reported in breast milk. The plasma elimination half-life for gentamicin has been reported to be 2 to 3 hours. Gentamicin does not appear to be metabolized and are excreted virtually unchanged in the urine by glomerular filtration.

Indication

Miramycin is indicated in the treatment of infections due to susceptible organisms in urinary tract infections, such as pyelonephritis, cystitis, and prostatitis, urethritis like gonorrhea or non-specific urethritis.

It is also indicated in respiratory tract infections like pneumonia, broncho-pneumonia, pleuritis, empyema, bronchitis, pharyngo-tonsillitis.

Other infections indicated include bacteremia meningitis, peritonitis, infected surgical wounds, and severe soft tissue infection including burns complicated by sepsis.

Recommended Dosage

Patients with normal renal function:

For I.M. use:

Urinary tract infections:

0.8 to 1.2 mg/kg/day in 2 to 3 equally divided doses for 7 to 10 days.

Other infections:

3 mg/kg/day in 2 to 3 equally divided doses for 7 to 10 days

Serious infections:

Up to 5 mg/kg/day in 3 equally divided doses for 7 to 10 days.

For intravenous use:

Dilute Miramycin into normal saline solution or dextrose 5% in water in the ratio of Miramycin 1 ml to the solution 50 to 500 ml and the solution should be infused over a period of 2 to 4 hours.

Route of Administration

Administered by intramuscular and intravenous.

Contraindication

Miramycin is contraindicated in patients with a history of hypersensitivity or toxic reaction to gentamicin. Contraindicated in patients with renal insufficiency or when hearing loss occurs.

Warning and Precautions

Use on patients with renal impairment should be done with caution, with the necessary adjustment of dosage and monitoring of blood levels. Prolonged use in pregnancy or infancy should be avoided to prevent damage to the eighth cranial nerve.

Interactions with other Medicaments

Not applicable.

Statement on usage during pregnancy and lactation

Pregnancy

Not applicable.

Lactation

Not applicable.

Adverse Effects/ Undesirable Effects

Allergic rash and skin reactions have occurred and local application had given rise to local allergy.

Overdose and Treatment

In the event of overdose or toxic reactions, hemodialysis may aid in the removal of the drug from the blood especially if renal function is or becomes compromised. Removal by peritoneal dialysis is at a much lower rate.

Storage Condition

Store below 30°C

Dosage forms and packaging available

40 mg/1ml (2ml) x 25 vials

Name and address of manufacturer

ATLANTIC LABORATORIES CORPORATION LTD.

111, Moo 7, Bang Phli Noi, Bang Bo, Samut Prakan 10560, Thailand.

Name and address of product registration holder

ATLANTIC LABORATORIES (M) SDN. BHD.

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